KP10 Network size, generalization and early stopping

> Level: third-year undergraduate | Self-study | English edition

Read this independently with weighted scores, basic differentiation and matrices as prerequisites. New terms are introduced when needed.

Open the matching slides

Learning objectives

- Count parameters and distinguish them from units.
- Select a checkpoint and distinguish three data roles.
- Explain why a cell bound is not a quality guarantee.

KP10 Network size, generalization and early stopping

One-sentence summary：Select size and training duration using held-out evidence, not training fit alone.

Plain explanation：Separate memorizing training answers from predicting new data before discussing size. Parameter counts below belong to explicit architectures, not hidden-unit counts alone.

Formal definition：The sections define terms, symbols and assumptions while keeping symbols consistent for the same quantity.

Example：All new worked examples are [Added] and checked in Python. Source experiment reports are identified separately.

Common misconception：Fewer parameters and longer fitting do not independently guarantee better new-data performance.

Practice：The final section provides questions, hints and worked checks for use after your own attempt.

What capacity can learn

Separate memorizing training answers from predicting new data before discussing size. Parameter counts below belong to explicit architectures, not hidden-unit counts alone.

Generalization: Correct prediction on data not used for fitting is generalization; memorizing training answers is different.

This continuous-prediction illustration exposes a trade-off also faced in classification. Find the main trend before treating sample fluctuations as rules.

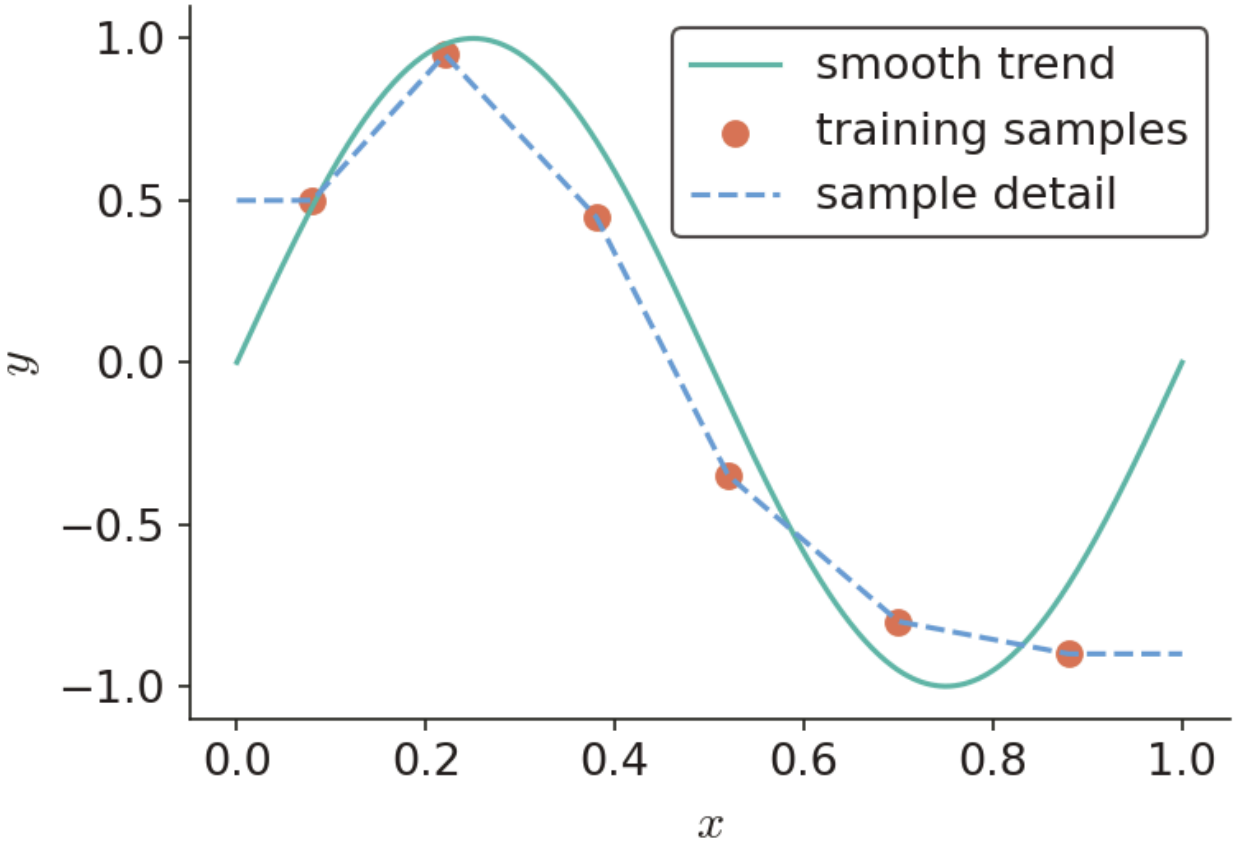


Figure `lec05-kp10-trend`：[Added] Compare the labelled alternatives, then check them against the numerical discussion. Schematic model drawings are not exact experimental traces.

Free parameter: A free parameter is an independently adjustable number; weights and biases both count.

Use two inputs, three hidden units and one output, with dense connections and one bias per computing unit.

Hidden layer: $(2 + 1) \times 3 = 9$. Output layer: $(3 + 1) \times 1 = 4$. There are $9 + 4 = 13$ parameters, not three.

Underfitting: Underfitting means the model misses important structure and can perform poorly even on training data.

Overfitting: Overfitting adapts to incidental training details at the expense of prediction on new data.

Size alone does not prove overfitting. Data, weight magnitudes, constraints and the fitting procedure also matter.

Choose checkpoints using held-out data

Even a fixed architecture may adapt too closely to training data. Examine two loss curves, choose among five checkpoints, and distinguish validation from testing.

Validation set: Validation data do not update weights; they help choose a model or training checkpoint.

Overtraining: Overtraining occurs when further fitting of a fixed architecture starts hurting validation performance.

Training loss keeps falling; validation loss rises after its minimum at epoch 3. The curves need not cross or start together.

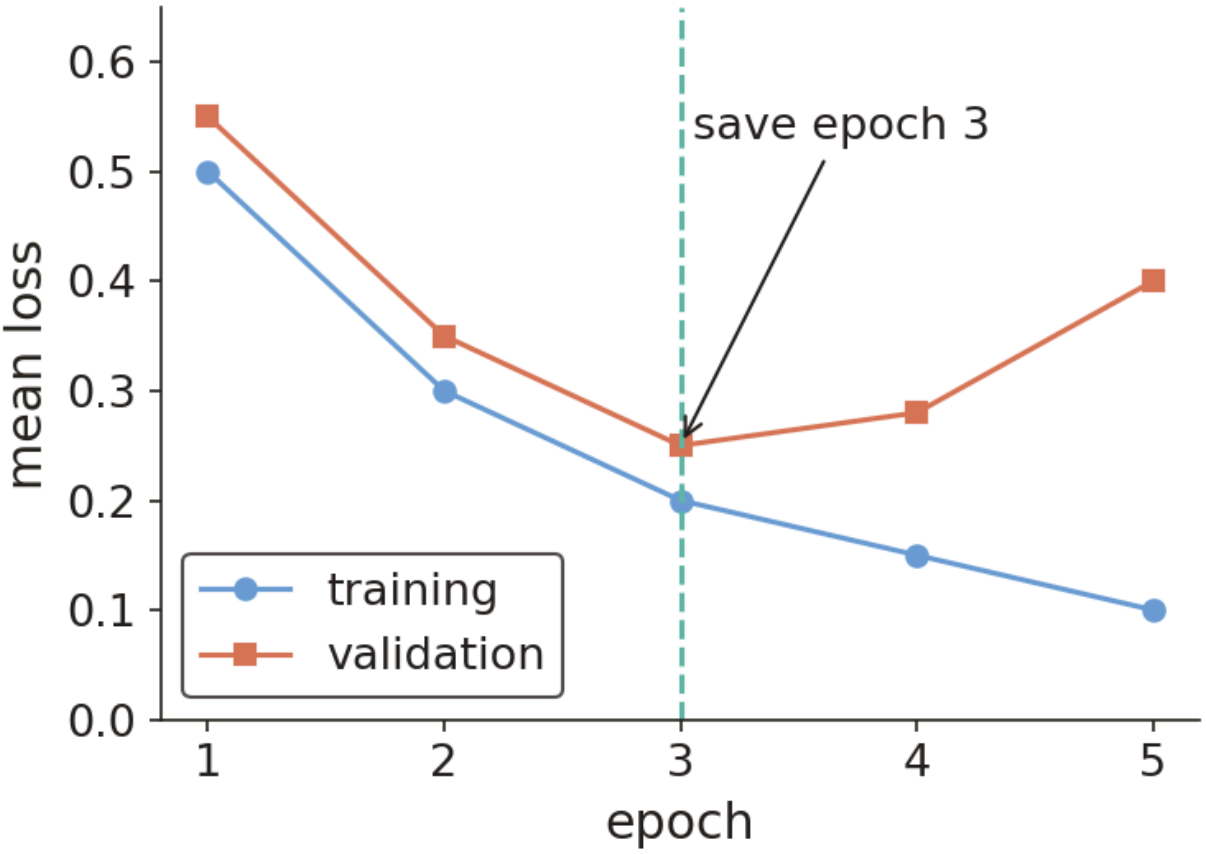


Figure `lec05-kp10-early`：[Added] Compare the labelled alternatives, then check them against the numerical discussion. Schematic model drawings are not exact experimental traces.

Early stopping: Early stopping uses validation performance and restores the best checkpoint; implementation may allow patience.

Validation losses at epochs 1–5 are $0.55, 0.35, 0.25, 0.28, 0.40$. The minimum is 0.25 at epoch 3.

Epoch 5 has lower training loss, but validation loss rises from 0.25 to 0.40: $0.40 - 0.25 = 0.15$.

Training data fit weights to patterns.

Validation data select size, penalty strength and checkpoints without directly fitting weights.

Test set: reserve it for evaluation after model selection. Reusing it to choose models breaks independence; avoid duplicate-subject and temporal leakage.

One error among ten gives $1/10 = 0.1$; two give $2/10 = 0.2$. One case changes the estimate by ten percentage points.

Repeated splits or cross-validation can support selection, but fitting and evaluation must remain separated. They do not create new independent data.

There is no best neuron count independent of the data and task.

Geometric capacity versus an actual XOR construction

Held-out selection asks about prediction, while geometry counts possible cells. XOR makes the distinction between a maximum and an actual arrangement concrete.

Cell-count bound: K hyperplanes partition d-dimensional space into at most M cells. General position attains the maximum; degenerate arrangements may not.

$M_{\max}(K, d) = \sum_{m=0}^{d} \binom{K}{m}$. K counts hidden units and d input dimensions; $\binom{K}{m} = 0$ when $m > K$.

General position excludes coincidences, parallelism and excessive intersections that reduce cell count; the precise condition depends on dimension.

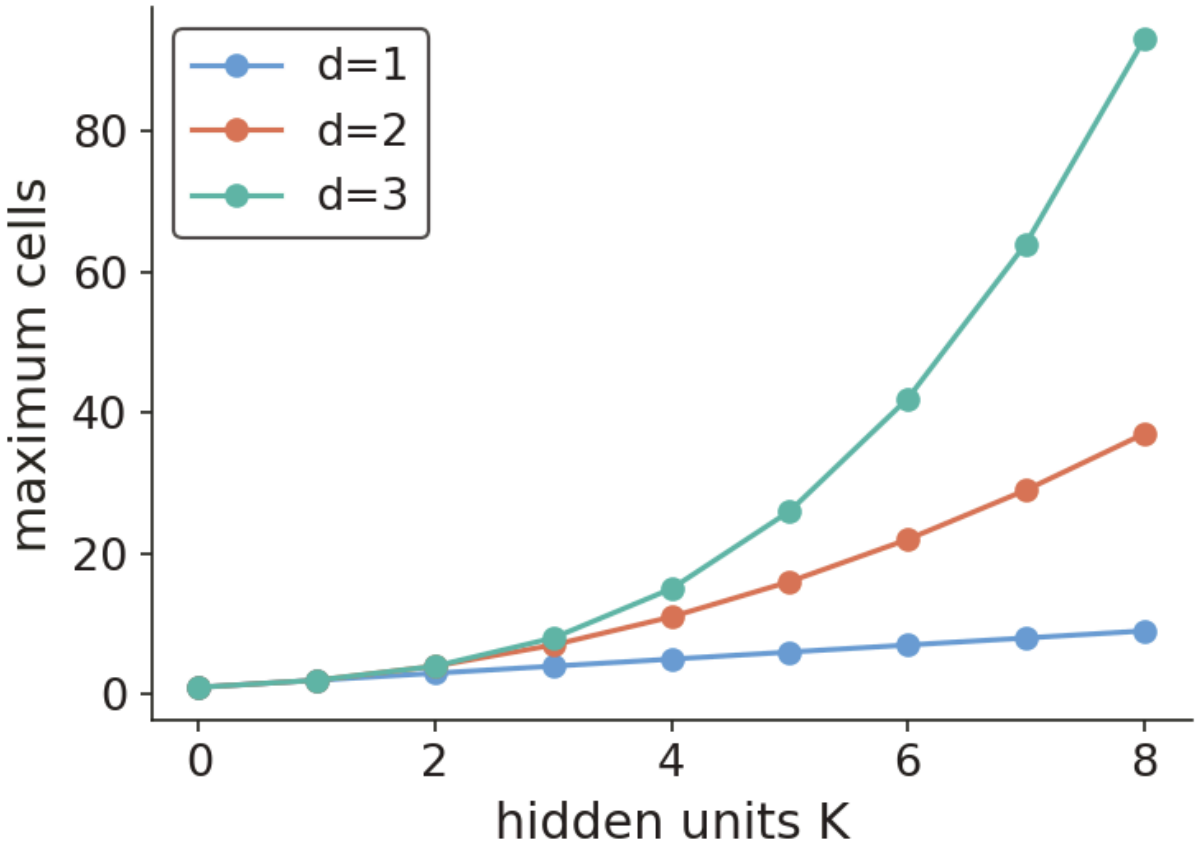


Figure `lec05-kp10-regions`：[Added] Compare the labelled alternatives, then check them against the numerical discussion. Schematic model drawings are not exact experimental traces.

For $d = 2, K = 2, M_{\max} = 1 + 2 + 1 = 4$. The three terms count choices of 0, 1 and 2 lines.

The XOR boundaries $x_1 + x_2 = 0.5$ and $x_1 + x_2 = 1.5$ are parallel: three actual bands, with both A points in the middle.

An upper bound of four accommodates three cells; sufficiency of two units still needs this explicit construction.

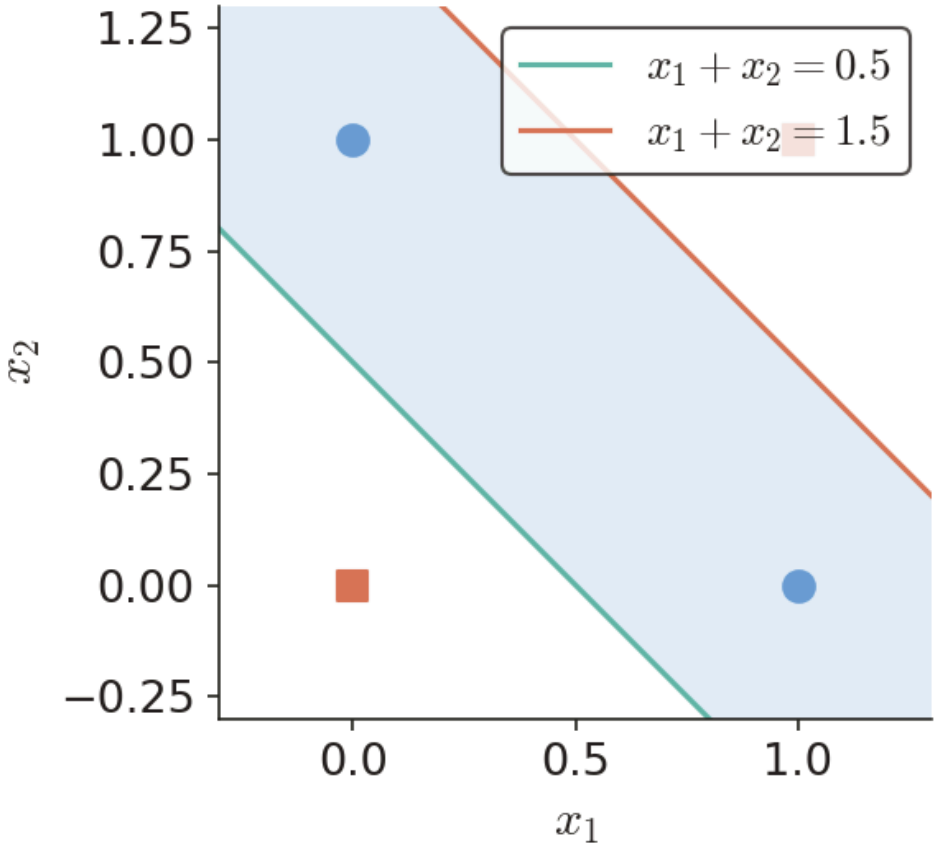


Figure `lec05-kp10-xor`：[Added] Compare the labelled alternatives, then check them against the numerical discussion. Schematic model drawings are not exact experimental traces.

Three size-selection strategies

A cell bound does not find good weights. Return to practical model search and compare shrinking a network with growing one.

Analytical methods estimate capacity using algebra or statistics, often without optimization details.

Pruning fits a larger network, then removes less useful connections by a rule.

Constructive methods grow units from a small architecture. All routes need validation; the next KP studies pruning.

Interpret the source demonstration within scope

With capacity and selection separated, the source sine demonstration becomes a case rather than a universal setting.

The source uses 15 training points and 10 independent test points for $y = \sin(2\pi x) + \epsilon$, where ϵ is observation noise.

$\hat{y} = \sum_{j=1}^{k} v_j \tanh(a_j x + b_j) + c$. Adjustable a, b, v and c give $3k + 1$ parameters; k counts hidden units.

The source describes k around 3–5 as useful and possible overfitting at k=15. These are not universal choices; this training was not rerun here.

Practice and worked checks

Cover the worked answers first, make both decisions, then check the numerical reasoning.

Question 1: training losses are $0.40, 0.20, 0.10$ and validation losses $0.45, 0.25, 0.30$. Choose a checkpoint by the validation minimum.

Question 2: count the maximum cells of three planar lines in general position using $\binom{3}{0} + \binom{3}{1} + \binom{3}{2}$. Does parallelism attain it?

1. The minimum of $0.45, 0.25, 0.30$ is 0.25: retain checkpoint 2, despite lower final training loss.
2. Three planar lines give $M_{\max} = 1 + 3 + 3 = 7$. Three parallel lines give only four bands.
3. Validation loss rises by $0.30 - 0.25 = 0.05$. Cell count did not enter the checkpoint comparison.

Why does the cell count have this form?

In general position the K-th hyperplane splits some existing cells. The additional pieces correspond to the cells formed by the previous K−1 hyperplanes inside this new, lower-dimensional plane. Hence

$$R(K, d) = R(K-1, d) + R(K-1, d-1).$$

Start from $R(0, d) = 1$ and $R(K, 0) = 1$. Pascal's identity gives $R(K, d) = \sum_{m=0}^{d} \binom{K}{m}$. This counts partitions; it does not ensure that the output layer separates arbitrary cell labels.

An executable checkpoint check

```
validation = [0.55, 0.35, 0.25, 0.28, 0.40]
best_epoch = min(range(len(validation)), key=validation.__getitem__) + 1
assert best_epoch == 3
```

This selects from supplied checkpoints, not trained networks. Real fitting also needs saved and restored weights, a fixed data split and a specified assessment metric.

What mean squared error measures

The source sine example predicts continuous values with $L = \frac{1}{N} \sum_{n=1}^{N} (\hat{y}_n - t_n)^2$, where t is target, hat y prediction and N the number of examples in that split. Errors one and minus one give $L = (1^2 + (-1)^2)/2 = 1$. Classification needs a threshold and an additional classification-error measure. The earlier five-checkpoint table is a teaching example, not an execution of the sine demo.

Glossary and symbol reminders

English	中文
Generalization	泛化
Free parameter	自由參數
Underfitting	欠擬合
Overfitting	過擬合
Validation set	驗證集
Overtraining	過度訓練
Early stopping	早停
Test set	測試集
Cell-count bound	區域數上界

Read symbols by their local definitions. Distinguish weights, scores, predictions and loss; identically named hyperparameters in different experiments may have different roles.

Sources and supplementary scope

- Chapter 4, KP10, accessed 2026-10-25.
- The source groups validation with testing, describes stopping at divergence and infers sufficiency from a cell bound. This material separates data roles, validation minima and constructive sufficiency. Real training losses also need not decrease monotonically.
- Source reference code: [Mirc 89].
- Original Figures 4.14–4.16 were inspected. New teaching diagrams do not claim to be the original figures or exact experimental curves.
- Historical reference codes follow the source. Full papers, theorem conditions and experimental records were not all available; those limits are retained.